

Computer Network and Communication CNS3200

Learning Outcome

- EXPLAIN INTERNETWORKING AND RELATED PROTOCOLS I.E UNICAST/MULTICAST (C3)
- CONNECT AND CONFIGURE LAB 2 (INTERNETWORKING)



Multicasting

- sending packet to addresses referring to group of hosts on one or more networks
 - multimedia “broadcast”
 - teleconferencing
 - database
 - distributed computing
 - real time workgroups
- have design issues in addressing / routing



LAN Multicast

- LAN multicast is easy
 - send to IEEE 802 multicast MAC address
 - since broadcast all stations will see packet
 - those in multicast group will accept it
 - only single copy of packet is needed
- but much harder in internetwork



Requirements for Multicasting

- router may have to forward more than one copy of packet
- need convention to identify multicast addresses (IPv4 Class D or IPv6 prefix)
- nodes translate between IP multicast addresses and list of networks containing group members
- router must translate between IP multicast address and network multicast address

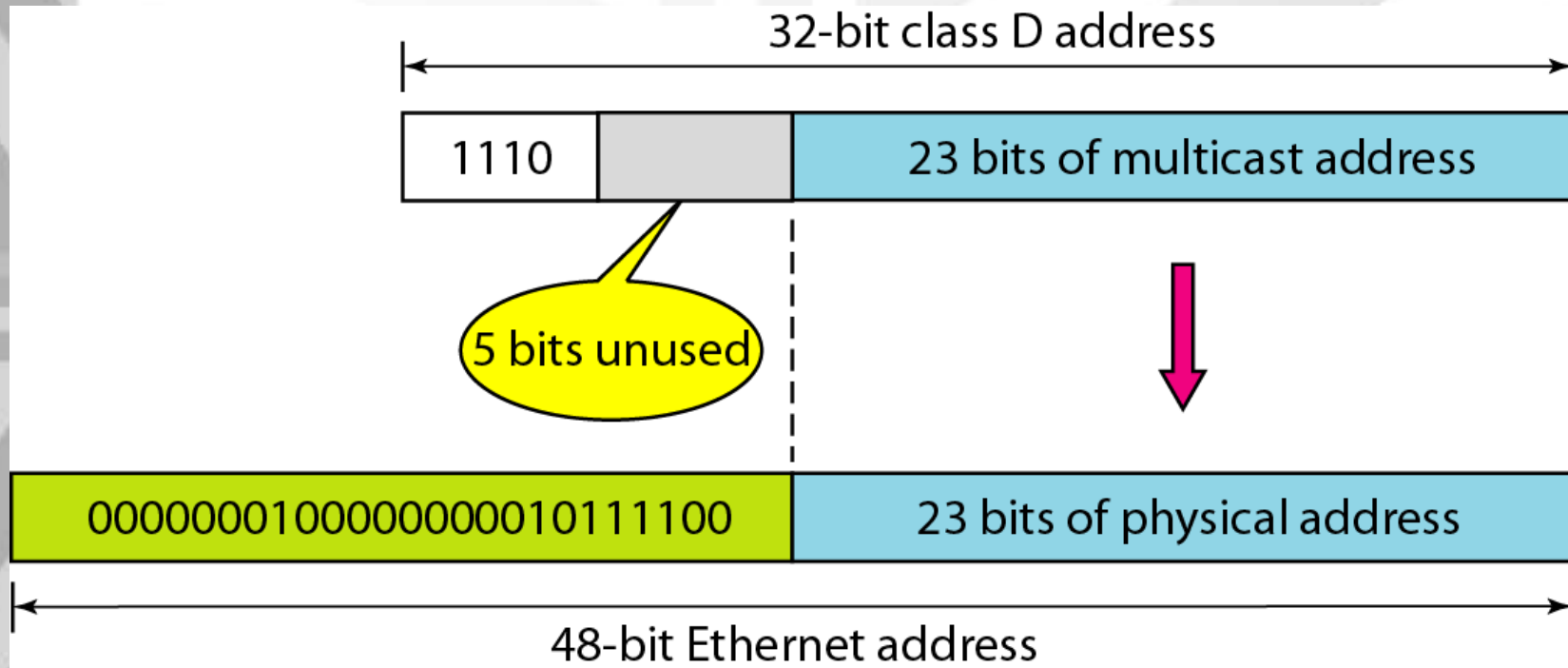


Requirements for Multicasting

- mechanism required for hosts to join and leave multicast group
- routers must exchange info
 - which networks include members of given group
 - sufficient info to work out shortest path to each network
- routing algorithm to work out shortest path
- routers must determine routing paths based on source and destination addresses



Figure 21.21 Mapping class D to Ethernet physical address



Note

An Ethernet multicast physical address is in the range
01:00:5E:00:00:00 to 01:00:5E:7F:FF:FF.



Change the multicast IP address 230.43.14.7 to an Ethernet multicast physical address.

Solution

We can do this in two steps:

- a. We write the rightmost 23 bits of the IP address in hexadecimal. This can be done by changing the rightmost 3 bytes to hexadecimal and then subtracting 8 from the leftmost digit if it is greater than or equal to 8. In our example, the result is 2B:0E:07.*



b. We add the result of part a to the starting Ethernet multicast address, which is 01:00:5E:00:00:00. The result is

01:00:5E:2B:0E:07

Change the multicast IP address 238.212.24.9 to an Ethernet multicast address.

Solution

- a. The rightmost 3 bytes in hexadecimal is D4:18:09. We need to subtract 8 from the leftmost digit, resulting in 54:18:09.*
- b. We add the result of part a to the Ethernet multicast starting address. The result is*

01:00:5E:54:18:09